



FRONTEND WORKSTREAM · CONSOLIDATED HANDOFF

Frontend Change Summary

A complete record of the judge-focused usability, information hierarchy, workflow, reporting, Proof Map, and navigation improvements made during the review.

32

affected frontend/status files,
including one new breadcrumb
component

342

repository tests passed; two
expected fixture-dependent skips

0

canonical analysis records,
dependencies, or backend
contracts deleted

Prepared: 24 September 2026

Audience: Judges, SOC analysts, cyber-security analysts, incident responders, and project reviewers

Scope: Frontend presentation and interaction only

EXECUTIVE SUMMARY

Outcome

The frontend was refined into a cleaner security-decision workflow. Judges can identify the highest-priority issue, understand its impact, trace supporting evidence, inspect affected sessions, review remediation, and export the report without first navigating dense identifiers or repeated metadata.

The route structure, Chain-of-Proof data contract, analysis records, backend adapters, and evidence-integrity boundaries were preserved. Technical detail was not deleted; lower-priority detail was moved into disclosures, inspectors, provenance sections, or the report appendix.

Important capability boundary: the backend-unavailable upload notice remains truthful. It appears when attempting unavailable file analysis, not on the prepared Overview. The interface was not changed to claim that an unavailable backend processed a capture.

1 Landing and analysis start

- The temporary judge-focused landing redesign was reverted, preserving the original capture-selection Start Analysis page.
- **Open Complete Demo** was renamed to **Review Available Analysis**.
- Distracting “demo” language was removed from the primary workflow while internal source validation remained intact.
- The unavailable-upload message now clearly states that no file was uploaded, stored, or analyzed.
- The prepared investigation remains reachable without submitting a capture.

2 Processing and completion

- Removed the simulated six-stage analysis sequence, fake percentages, and “Simulating” status labels.
- Replaced them with a concise analysis-preparation state shared by prepared and backend result flows.
- The completion page now guides users through Overview, Findings, Proof Map, Recommendations, and Report.
- Capture filename fallback now comes from the supplied analysis record when no local filename exists.
- Removed source-specific labels such as “Complete sample investigation” and “Production API” from the main presentation.

3 Overview dashboard

- Added a decision-focused lead section for the highest-priority deterministic finding.
- Added impact and direct navigation to supporting evidence, session investigation, and remediation.
- Reduced competing summary cards and introduced four concise decision metrics.
- Kept **Policy Risk** and **ML Anomaly** visibly separate.
- Added prioritized sessions so analysts immediately see where attention is required.

- Added an Assessment Coverage matrix distinguishing assessed, observed, unavailable, not observed, and not assessed states.
- Simplified protocol and TLS-outcome visualizations and improved card spacing, responsive behavior, severity treatment, and hierarchy.

4 Sessions

- Simplified the introduction and removed repeated analysis metadata.
- Reduced and compacted summary metrics so the session inventory appears sooner.
- Prioritized readable protocol, stream, endpoint, transition, TLS, evidence, finding, and limitation information.
- Removed raw session IDs from the primary visual hierarchy while keeping them available in technical views.
- Placed optional search and filtering controls inside an accessible disclosure.
- Improved both desktop table spacing and mobile session cards.

5 Session X-Ray

- Shortened the introduction and brought the observed session timeline forward.
- Moved the generic STARTTLS sequence into a collapsed reference guide.
- Reorganized crypto data into TLS outcome, negotiated parameters, other observations, certificate observations, certificate assessment, and limitations.
- Simplified linked findings while preserving evidence, limitations, Policy Risk, and ML Anomaly separation.
- Replaced the misleading disabled scope dropdown with a clear current-session indicator and node/relationship counts.

6 Findings and ML results

- Presented deterministic policy findings first as security decisions with severity, affected scope, rationale, and impact.
- Moved rule internals, raw records, fact references, and technical identifiers into disclosures.
- Placed optional filtering controls behind an accessible disclosure.
- Separated Policy Findings from ML Anomaly results and improved empty, unavailable, and not-run states.
- Prevented ML outputs from being presented as deterministic policy conclusions.

7 Proof Map

- Moved the evidence graph above its explanatory legend.
- Converted the large legend into a collapsed "How to read paths and uncertainty states" section.
- Simplified the incomplete-evidence warning and compacted summary metrics.
- Preserved every contract-declared relationship and maintained exact support-path behavior.
- Kept the standalone session selector functional while showing a fixed current-session state inside Session X-Ray.
- Verified scope changes, finding highlighting, filtering, fit controls, inspectors, focus behavior, and mobile interaction.

8 Session comparison

- Changed the workspace to be difference-first and hide matching fields by default.
- Added a concise difference count and readable labels for transitions, TLS, findings, anomalies, evidence, and limitations.
- Improved posture summaries for both selected sessions.
- Preserved exact technical values in expandable detail.

9 Recommendations

- Refocused the page around what to fix, why it matters, and how to verify the change.
- Kept policy remediation and ML-related guidance separate.
- Added URL-backed tabs so direct links open the intended recommendation view.
- Removed finding/session identifiers from the main cards but retained them in technical details.
- Improved grouping of linked findings, affected sessions, remediation steps, verification steps, and evidence.

10 Report and export

The report now follows the review order below:

1. Executive summary
 2. Findings and affected sessions
 3. Recommendations and verification
 4. Policy Risk and separate ML status
 5. Scope and limitations
 6. Capture and analysis provenance
 7. Evidence and technical records
- Added an executive finding with impact and direct evidence references.
 - Moved hashes, IDs, versions, raw scores, thresholds, contributions, manifests, and source records into technical disclosures.
 - Renamed the export action to **Download Chain JSON**.
 - Renamed **Print / Save as PDF** to **Print report** and clarified that Save as PDF is a browser capability.
 - Changed the local filename from `analysis-id.demo-chain.json` to `analysis-id.chain.json`.
 - Removed the sample-data prefix from the browser report title.

11 Presentation cleanup

Removed from primary presentation

- Repeated dataset banners
- "Static sample" labels
- "Complete demo" wording
- "Local walkthrough" and simulation labels
- Source-specific report titles
- Demo-specific export text and filenames
- Repeated notices and metadata blocks

Preserved internally

- Validated local Chain dataset
- Prototype-ID source routing
- Runtime and integrity checks
- Evidence boundaries and limitations
- Backend-unavailable safeguards
- Protection against silent fallback
- Complete technical source records

12 Navigation

- Added an accessible desktop sidebar hide/show control similar to ChatGPT.
- Preserved the existing mobile navigation.
- Added route-aware breadcrumbs for Overview, Sessions, Session X-Ray, Compare, Findings, Proof Map, Recommendations, and Report.
- Made parent breadcrumb items navigable and ensured narrow-screen truncation.

Data removed versus presentation removed

No canonical analysis data was deleted. The work changed presentation priority, progressive disclosure, wording, and navigation—not the underlying Chain records.

Preserved data	Presentation treatment
Captures, sessions, protocol events and TLS observations	Summarized first; complete records remain available.
Certificate observations, facts, findings and recommendations	Readable decisions first; exact records in technical details.
Evidence references and limitations	Kept explicit and linked to relevant findings and sessions.
Policy scores and ML results	Kept separate and independently labelled.
IDs, hashes, thresholds, versions and manifests	Removed from the default visual layer, not deleted.
Provenance and integrity metadata	Retained in report provenance and appendices.

Structural boundaries preserved

- Routes and major application structure remain unchanged.
- The Chain remains the report and dashboard source of truth.
- Analysis data contracts and backend adapters remain intact.
- Policy Risk and ML Anomaly remain separate outputs.
- No authentication, SIEM integration, live capture, or product-generated PDF system was added.
- No dependency, backend implementation, controlled evidence, or fixture was changed.

Verification completed

Automated checks

- TypeScript passed
- ESLint passed
- Production build passed for all routes
- Dependency audit: zero vulnerabilities
- Proof Map regression and fit checks passed
- Report consistency checks passed
- Ruff and formatting checks passed
- `git diff --check` passed

Runtime review

- 342 repository tests passed
- 2 expected skips: frozen T01 PCAP absent
- Desktop judge journey inspected
- 500 px responsive journey inspected
- Sidebar and breadcrumbs tested in Chrome
- Proof Map scope behavior tested live

Final repository state

The accumulated frontend work contains 31 modified tracked files plus one new breadcrumb component. No commit, push, deployment, backend mutation, fixture regeneration, evidence rewrite, or dependency addition was performed.

This document summarizes the verified frontend working-tree changes as of 24 September 2026. It does not claim implementation of capabilities that are absent from the backend.